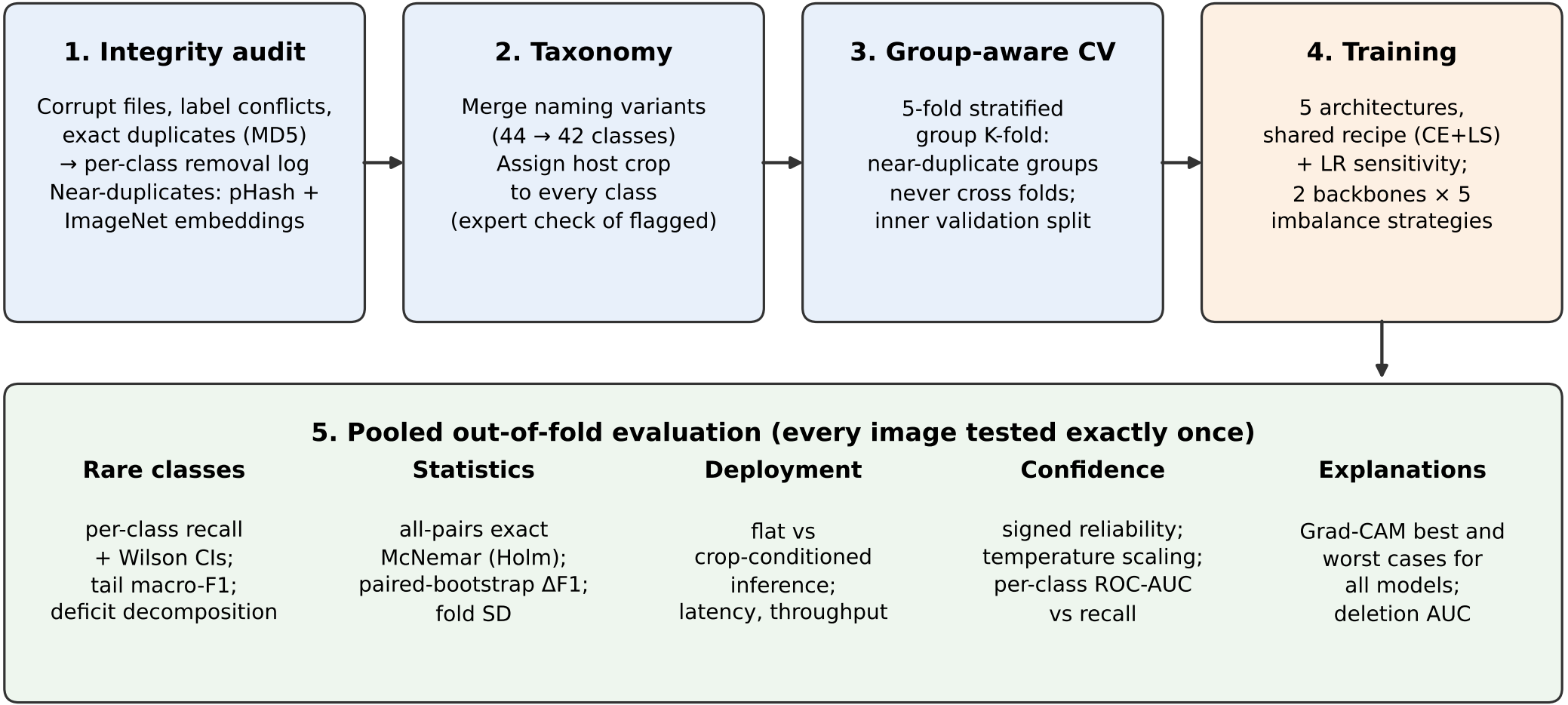
1. Integrity audit
Corrupt files, label conflicts, exact duplicates (MD5) → per-class removal log
Near-duplicates: pHash + ImageNet embeddings
2. Taxonomy
Merge naming variants (44 → 42 classes)
Assign host crop to every class
(expert check of flagged)
3. Group-aware CV
5-fold stratified group K-fold:
near-duplicate groups never cross folds;
inner validation split
4. Training
5 architectures, shared recipe (CE+LS) + LR sensitivity;
2 backbones × 5 imbalance strategies
5. Pooled out-of-fold evaluation (every image tested exactly once)
Rare classes
per-class recall + Wilson CIs; tail macro-F1; deficit decomposition
Statistics
all-pairs exact McNemar (Holm); paired-bootstrap ΔF1; fold SD
Deployment
flat vs crop-conditioned inference; latency, throughput
Confidence
signed reliability; temperature scaling; per-class ROC-AUC vs recall
Explanations
Grad-CAM best and worst cases for all models; deletion AUC